

ShopFlow AWS Project

Secrets Manager Design

Encrypted Secrets - Auto-Generated Passwords - KMS

Module 4 of 18

Field	Value
Version	1.0
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Region	ap-south-1 (Mumbai)
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Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

AWS Secrets Manager stores sensitive credentials encrypted at rest using AWS KMS. Application containers retrieve secrets via IAM roles at runtime -- the secret value never appears in source code, CloudFormation templates, or environment variable files in plain text. ShopFlow stores five secrets covering all credential categories required by ShopVerse.

- Passwords auto-generated by Secrets Manager -- developer never chooses or sees them
- ECS task definitions inject secrets as environment variables at container startup
- Supports automatic rotation -- DB password updated in both AWS and MySQL atomically
- Cost: \$0.40 per secret/month + \$0.05 per 10,000 API calls = ~\$2/month total

2. Secrets Inventory

Secret Name (Path)	Type	What It Stores	Used By
/shopflow/db/master	Auto-generated JSON	RDS admin username + random 32-char password	RDS CloudFormation stack
/shopflow/db/app	Auto-generated JSON	App DB username + random 32-char password	ECS task (DB_USER, DB_PASS env vars)
/shopflow/app/jwt-secret	Auto-generated JSON	64-char HS512 JWT signing key	ECS task (JWT_SECRET env var)
/shopflow/app/redis-auth	Auto-generated JSON	32-char alphanumeric Redis AUTH token	ECS task + ElastiCache cluster
/shopflow/app/third-party	Manual JSON	Stripe API key, SendGrid API key (placeholder)	Lambda email/payment functions

3. Secret Value Format

3.1 Database Master Secret (auto-generated at deploy time)

```
{ "username": "shopflow_admin", "password": "<32-char random>" }
```

3.2 JWT Secret (64-char alphanumeric, no special chars)

```
{ "jwt_secret": "<64-char alphanumeric string>" }
```

3.3 Redis AUTH Token (ElastiCache requires 16-128 alphanumeric chars)

```
{ "auth_token": "<32-char alphanumeric string>" }
```

4. How ECS Reads Secrets

ECS task definitions reference the secret ARN. ECS calls GetSecretValue at task startup, parses the JSON key, and injects the value as an environment variable.

Environment Variable	Source Secret	JSON Key
DB_USER	/shopflow/db/app	username
DB_PASS	/shopflow/db/app	password

Environment Variable	Source Secret	JSON Key
JWT_SECRET	/shopflow/app/jwt-secret	jwt_secret
REDIS_AUTH_TOKEN	/shopflow/app/redis-auth	auth_token

5. Secrets Manager vs SSM Parameter Store

Attribute	Secrets Manager	SSM Parameter Store
Use case	Passwords, tokens, API keys	Endpoints, feature flags, config
Encryption	Always (KMS)	Optional (SecureString)
Rotation	Yes (automated with Lambda)	No
Cost	\$0.40/secret/month	Free (Standard)
ShopFlow use	DB passwords, JWT, Redis auth	Hostnames, ports, feature flags